RQ Answers

omitted

2024-04-05

```
glimpse(all_data)
```

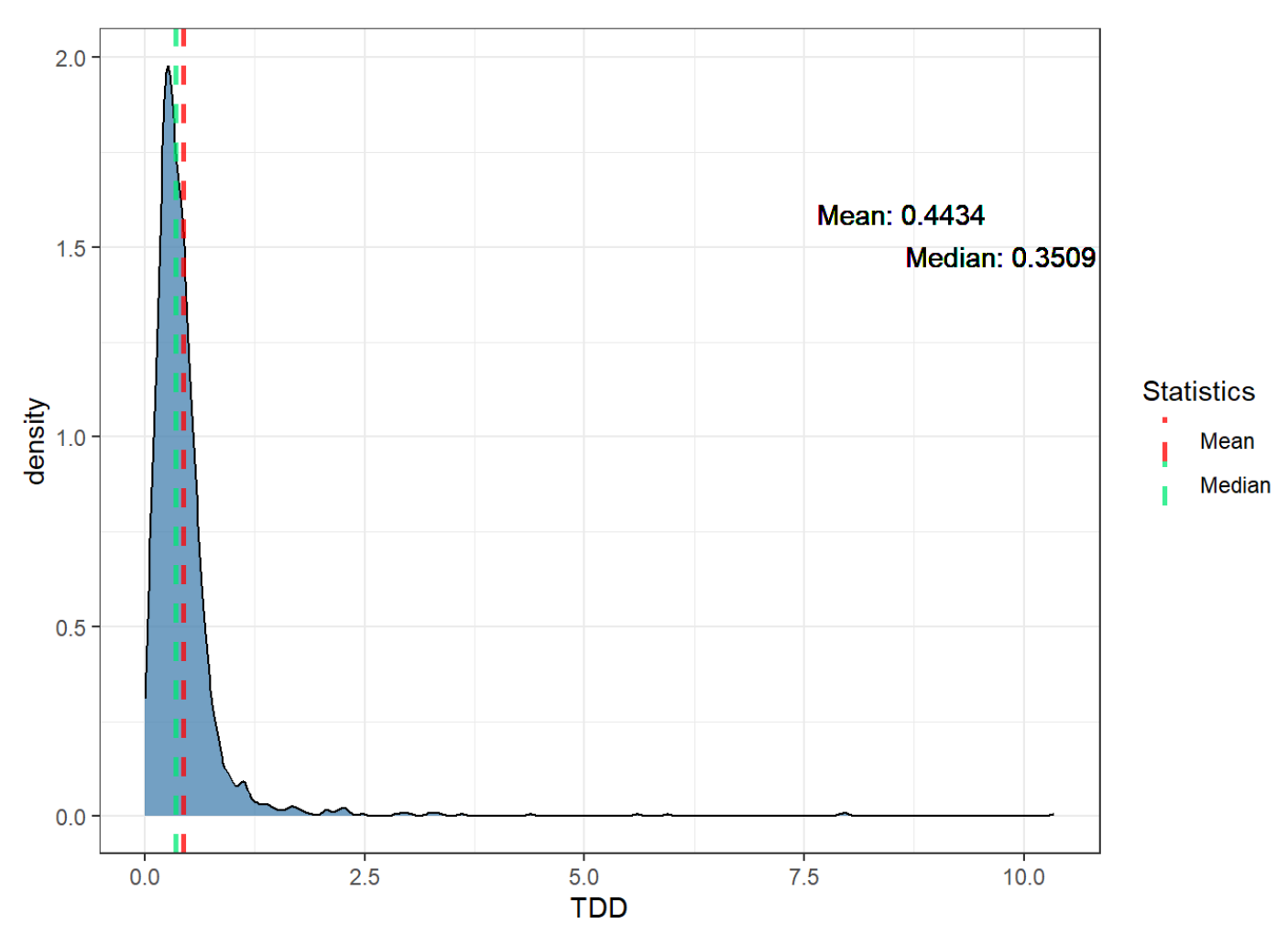
```
## Rows: 175,667
## Columns: 11
## $ repo              <chr> "accumulo", "accumulo", "accumulo", "accumulo", "a…
## $ pr_number         <int> 1302, 1302, 1302, 1302, 1302, 1302, 13…
## $ rule              <chr> "java:S1192", "java:S3776", "java:S1319", "java:S1…
## $ severity          <chr> "CRITICAL", "CRITICAL", "MINOR", "CRITICAL", "CRIT…
## $ file              <chr> "core/src/main/java/org/apache/accumulo/core/metad…
## $ type              <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status            <chr> "OPEN", "OPEN", "OPEN", "OPEN", "CLOSED", "OPEN", …
## $ debt              <int> 8, 6, 10, 5, 6, 2, 5, 15, 30, 109, 30, 30, 5, 5, 5…
## $ ncloc_affected_file <int> 347, 347, 347, 347, 347, 347, 347, 347, 252, 252, …
## $ complexity        <int> 68, 68, 68, 68, 68, 68, 68, 68, 86, 86, 86, 86, 86…
## $ origin            <chr> "NEW", "NEW", "NEW", "NEW", "PRE-EXISTING", "PRE-E…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 1569 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 1569 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

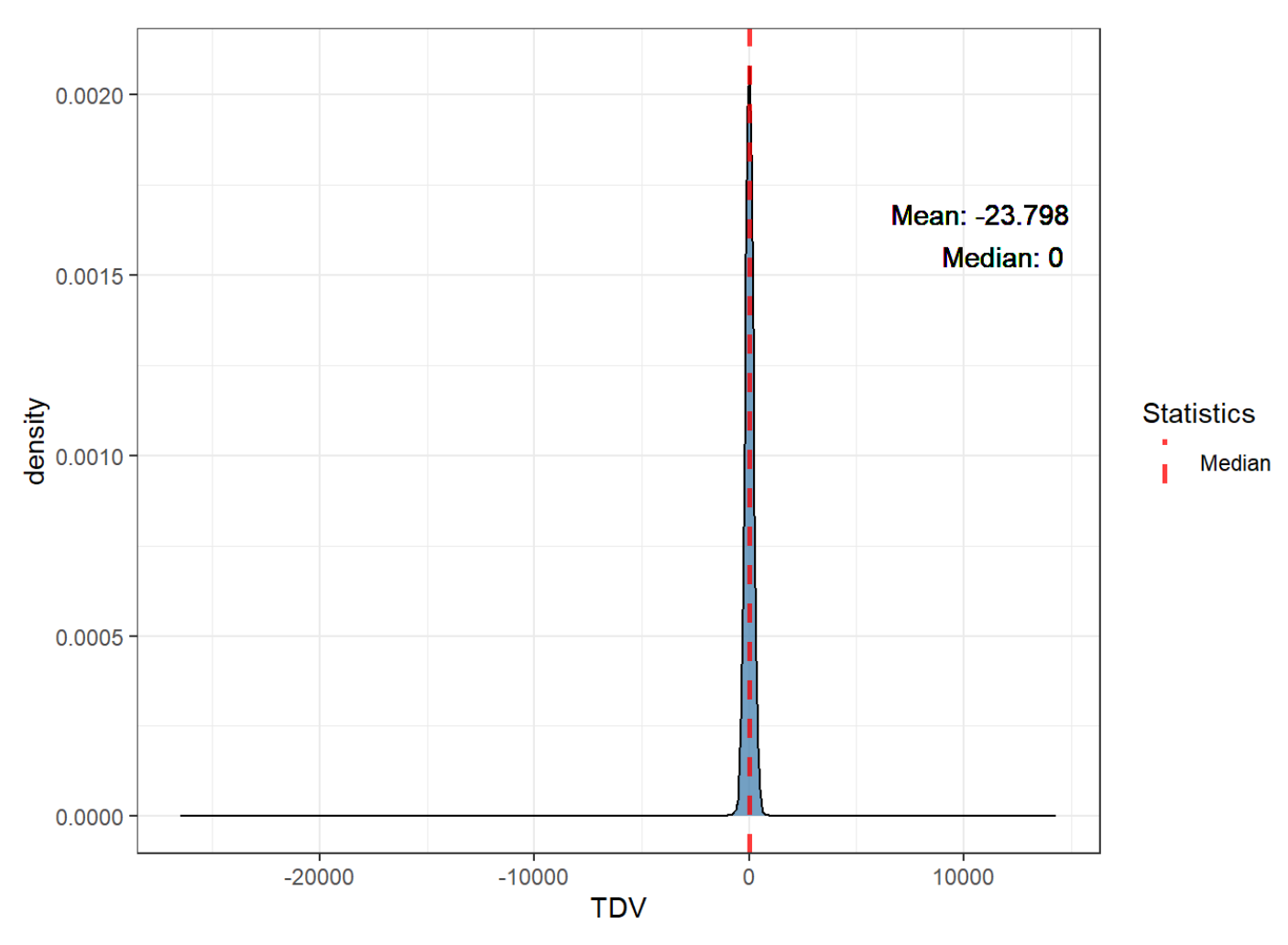


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 1569 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 1569 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 12 × 4
##    repo              tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##    <chr>                           <dbl>             <dbl>                 <dbl>
##  1 accumulo                        26.5              53.4                  20.1
##  2 cayenne                         12.8              72.3                  14.9
##  3 commons-collecti…               35.3              64.7                   0
##  4 commons-io                      22                72                     6
##  5 commons-lang                     9.68             89.2                   1.08
##  6 helix                           20                44.9                  35.1
##  7 httpcomponents-c…               16.4              61.8                  21.8
##  8 maven-surefire                  20                80                     0
##  9 opennlp                         44.6              44.6                  10.8
## 10 struts                          35.2              44.3                  20.5
## 11 wicket                          15.6              59.4                  25
## 12 zookeeper                       14.3              71.4                  14.3
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                      <dbl>
## 1                     22.7                   63.2                       14.1
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 12 × 3
##    repo                   preexisting_tdv_equal_to_zero preexisting_tdv_greater…¹
##    <chr>                                          <dbl>                     <dbl>
##  1 accumulo                                        62.7                     37.3
##  2 cayenne                                         80.8                     19.2
##  3 commons-collections                             58.8                     41.2
##  4 commons-io                                      76                       24
##  5 commons-lang                                    90.3                      9.68
##  6 helix                                           62.7                     37.3
##  7 httpcomponents-client                           75.7                     24.3
##  8 maven-surefire                                  80                       20
##  9 opennlp                                         42.5                     57.5
## 10 struts                                          55.4                     44.6
## 11 wicket                                          61.3                     38.7
## 12 zookeeper                                       78.6                     21.4
## # ℹ abbreviated name: ¹preexisting_tdv_greater_than_zero
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                          68.7                                   31.3
```

Percentages for new TDV by repo

```
## # A tibble: 12 × 3
## # Groups:   repo [12]
##    repo                   new_tdv_equal_to_zero new_tdv_greater_than_zero
##    <chr>                                  <dbl>                     <dbl>
##  1 accumulo                                73.2                      26.8
##  2 cayenne                                 90                        10
##  3 commons-collections                      0                       100
##  4 commons-io                              77.8                      22.2
##  5 commons-lang                           100                         0
##  6 helix                                   72.4                      27.6
##  7 httpcomponents-client                   93.1                       6.9
##  8 maven-surefire                         100                         0
##  9 opennlp                                 96.8                       3.23
## 10 struts                                  85.7                      14.3
## 11 wicket                                  88.9                      11.1
## 12 zookeeper                               85.7                      14.3
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                       80.3                           19.7
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositiries.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 × 13
## # Groups:   rank [10]
##     rank accumulo      cayenne `commons-collections` `commons-io` `commons-lang`
##    <int> <chr>         <chr>   <chr>                 <chr>        <chr>
## 1      1 java:S117  -… java:S… java:S1125  -  8      java:S1125 … java:S3358  -…
## 2      2 java:S2589  … java:S… java:S1197  -  5      java:S1192 … java:S2293  -…
## 3      3 java:S112   … java:S… java:S1612  -  2      java:S1612 … java:S1155  -…
## 4      4 java:S1125  … java:S… java:S1128  -  1      java:S1135 … java:S2326  -…
## 5      5 java:S101   -… java:S… java:S1192  -  1      java:S1066 … java:S1066  -…
## 6      6 java:S2293  … java:S… java:S2583  -  1      java:S3008 … java:S117  - …
## 7      7 java:S3776  … java:S… java:S1155  -  1      java:S1128 … java:S1125  -…
## 8      8 java:S1155  … java:S… java:S1602  -  1      java:S3551 … java:S1854  -…
## 9      9 java:S1192  … java:S… <NA>                  java:S1133 … java:S1126  -…
## 10    10 java:S100   -… java:S… <NA>                  java:S1874 … java:S1135  -…
## # ℹ 7 more variables: helix <chr>, `httpcomponents-client` <chr>,
## #   `maven-surefire` <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## #   zookeeper <chr>
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 57 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S3776             82
##  2 java:S1192             91
##  3 java:S2293             92
##  4 java:S1125            100
##  5 java:S1161            102
##  6 java:S1135            106
##  7 java:S3740            107
##  8 java:S1874            108
##  9 java:S1066            111
## 10 java:S117             111
## # ℹ 47 more rows
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 13
## # Groups:   rank [10]
##     rank accumulo      cayenne `commons-collections` `commons-io` `commons-lang`
##    <int> <chr>         <chr>   <chr>                 <chr>        <chr>
## 1      1 java:S117  -… java:S… java:S1125  -  87     java:S1133 … java:S1168  -…
## 2      2 java:S2589  … java:S… java:S2160  -  52     java:S1192 … java:S1133  -…
## 3      3 java:S101  -… java:S… java:S128  -  52      java:S3776 … java:S3776  -…
## 4      4 java:S1125  … java:S… java:S2065  -  51     java:S127  … java:S1149  -…
## 5      5 java:S3776  … java:S… java:S3776  -  46     java:S1135 … java:S1905  -…
## 6      6 java:S112  -… java:S… java:S1192  -  41     java:S1130 … java:S1192  -…
## 7      7 java:S1192  … java:S… java:S131  -  30      java:S4042 … java:S127  - …
## 8      8 java:S2293  … java:S… java:S1948  -  20     java:S2183 … java:S135  - …
## 9      9 java:S116  -… java:S… java:S1452  -  20     java:S1125 … java:S1118  -…
## 10    10 java:S2142  … java:S… java:S2259  -  19     java:S3626 … java:S2259  -…
## # ℹ 7 more variables: helix <chr>, `httpcomponents-client` <chr>,
## #   `maven-surefire` <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## #   zookeeper <chr>
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 51 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S3776             44
##  2 java:S1133             70
##  3 java:S1192             77
##  4 java:S1125            103
##  5 java:S1135            103
##  6 java:S1186            109
##  7 java:S1117            111
##  8 java:S117             112
##  9 java:S1874            112
## 10 java:S3740            112
## # ℹ 41 more rows
```

Outliers

Higher TDD

```
## # A tibble: 1,569 × 5
##    repo          pr_number total_debt total_ncloc   tdd
##    <chr>             <int>      <int>       <int> <dbl>
##  1 wicket              676        600          58 10.3
##  2 wicket              708       1996         250  7.98
##  3 helix              1975        261          33  7.91
##  4 accumulo           2922        410          69  5.94
##  5 accumulo           4215        140          25  5.6
##  6 opennlp             561      74742       17056  4.38
##  7 wicket              351       2159         599  3.60
##  8 accumulo           3612        724         215  3.37
##  9 commons-lang        333        470         143  3.29
## 10 wicket              495       2125         660  3.22
## # ℹ 1,559 more rows
```

Higher TDV

```
## # A tibble: 1,569 × 3
##    pr_number repo       tdv
##        <int> <chr>    <int>
## 1       2022 accumulo 14300
## 2       1946 accumulo  3875
## 3       2335 accumulo   935
## 4       1891 accumulo   643
## 5       4054 accumulo   573
## 6       1605 accumulo   508
## 7        522 cayenne    446
## 8       3827 accumulo   422
## 9       2249 helix      397
## 10      3563 accumulo   392
## # ℹ 1,559 more rows
```

Lower TDV

```
## # A tibble: 1,569 × 3
##    pr_number repo        tdv
##        <int> <chr>     <int>
## 1        799 struts   -26485
## 2        561 opennlp  -12880
## 3       3604 accumulo  -5740
## 4       3402 accumulo  -4826
## 5        670 struts    -1113
## 6       3229 accumulo   -726
## 7       2705 accumulo   -724
## 8       3117 accumulo   -691
## 9       2966 accumulo   -600
## 10      3327 accumulo   -508
## # ℹ 1,559 more rows
```

Extra

Number of PRs with pre-existing TD

```
##      n
## 1 1554
```

Number of java:S1192 issues that affect test code

```
##   n
## 1 0
```

Number of java:S1192 issues that affect not test code

```
##      n
## 1 8262
```

Summary NCLOC

```
##      ncloc
##  Min.   :     3
##  1st Qu.:   226
##  Median :   566
##  Mean   :  2120
##  3rd Qu.:  1479
##  Max.   :202678
```